

M. ZARYAB MALIK

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SUMMARY

BS Computing, Mathematics & AI student who independently architects and ships production AI systems end-to-end — RAG chatbots, face-recognition pipelines, and campus automation platforms — spanning backend, database, deployment, and Android packaging. Also freelances in graphic design and brand identity, delivering packaging, listing graphics, and brand identity work for e-commerce clients.

TECHNICAL SKILLS

- **Languages:** Python, C++, JavaScript, HTML/CSS, SQL
- **AI/ML:** Retrieval-Augmented Generation (RAG), LLM integration (Groq, Gemini), face recognition (FaceNet, ArcFace, MediaPipe), vector search (ChromaDB, FAISS), prompt engineering
- **Backend & Data:** FastAPI, PostgreSQL, Firebase, SentenceTransformers
- **Deployment:** Railway, Netlify, GitHub Pages, PWA Builder (Android APK packaging)
- **Graphic Design:** Adobe Photoshop, Adobe Illustrator, Canva, brand identity design, product packaging design, Amazon listing graphics, social media creatives, typography & layout
- **Tools:** Git/GitHub, VS Code, Cursor

PROJECTS

LGU Chatbot — RAG-based University Assistant

- Architected and deployed a retrieval-augmented generation chatbot end-to-end using FastAPI, ChromaDB, SentenceTransformers, and Groq's LLM API
- Engineered Firebase authentication, persistent chat history, and a CrossEncoder reranker that improved retrieval relevance over baseline vector-similarity search
- Shipped as an installable Android APK via PWA Builder, with backend on Railway and frontend on Netlify

LGU Gatepass — Smart Campus Exit Management System

- Independently architected a gate pass management system automating student exit verification through a multi-step HOD approval workflow
- Built the full stack solo — Python/FastAPI backend, PostgreSQL database, face-recognition verification, and an admin dashboard for approve/reject decisions
- Deployed on Railway and Netlify; packaged as an installable Android APK

GPS + Face Recognition Attendance System — Campus Attendance Automation

- Engineered a geofenced attendance system combining polygon-based GPS verification with real-time face recognition (FastAPI, ArcFace, FAISS)
- Redesigned the geofence from a circular to a polygon boundary and added a live GPS position indicator on an SVG campus map for accuracy
- Deployed on Railway; packaged as a TWA Android APK

FREELANCE GRAPHIC DESIGN

- Delivered product packaging, Amazon listing graphics, and brand identity design (logos, color systems, style guides) for e-commerce clients via Fiverr
- Designed social media creatives and marketing collateral using Adobe Photoshop, Adobe Illustrator, and Canva
- Managed client revisions end-to-end — concept, drafts, and final print/web-ready deliverables

EDUCATION

